Sreenath Karthick Muthuvel

Ernakulam, Kerala — msreenathkarthick2005@gmail.com — +91-9895819812 — LinkedIn — GitHub — Portfolio

SUMMARY

Computer Science undergraduate at IIT Palakkad with experience in programming languages, formal verification, AI systems, and software engineering. Interested in applied research at the intersection of programming languages, software engineering, formal methods, and AI.

EDUCATION

Indian Institute of Technology Palakkad *Aug 2023 - Apr 2027*
B.Tech in Computer Science and Engineering **CGPA: 9.55/10.0**
TOEFL iBT: 110/120 (**CEFR: C1**)
GRE: 317.5/346

EXPERIENCE

Research Intern *May 2025 – Jul 2025*
Defence Research and Development Organisation (DRDO: DYSL-CT), Chennai

- Researched and implemented Explainable AI techniques including **LIME, SHAP, and Class Activation Maps** to interpret deep-learning model predictions.
- Developed **PeliCAM**, a Python/PyQt application using **PyTorch** for model interpretability, supporting CAM analysis, LIME explanations, saliency visualization, and debug predictions across 3+ neural-network models.
- Contributed to text-classification systems for English and Indian languages.

Software Intern *Feb 2025 – Dec 2025*
Learning Sciences Research Group, Homi Bhabha Centre for Science Education (HBCSE – TIFR)

- Developed interactive web-based tools using **JavaScript, HTML, and CSS** for mathematical learning and experimentation.
- Contributed to a research project analyzing the learning processes of **100+ students**.
- Co-authored a peer-reviewed paper published in the *Proceedings of the 33rd International Conference on Computers in Education (ICCE 2025).*

Custom Software Engineering Associate *May 2026 – Jul 2026*
Accenture – Coimbatore

- Gained hands-on experience with Microsoft Azure security, including Virtual Machines, Network Security Groups, RBAC, Azure Policies, and Log Analytics.
- Worked with Azure monitoring and security tools including **Azure Monitor** and **Microsoft Defender for Cloud**.
- Performed security-log analysis using **Kusto Query Language (KQL)**.

PROJECTS

Tiger Compiler - Compiler for a Subset of the Tiger Language *Course Project*

- Developed a compiler for a subset of the Tiger programming language in **OCaml**, supporting assignments, arithmetic expressions, `print`, `for`, `if-then`, and `if-then-else` constructs.
- Implemented the compilation pipeline from lexical analysis and parsing through AST construction, Tree IR, canonicalization, basic-block formation, and MIPS code generation.
- Built compiler infrastructure for control-flow analysis, including basic-block construction and graph modules.
- Implemented naive register allocation with spilling and generated executable MIPS assembly compatible with the SPIM simulator.
- *Tools:* OCaml, Menhir, OCamllex, MIPS Assembly, SPIM.
- GitLab Repository

Formal Verified Tool Chain *Open Ended Lab Project*

- Developing a software toolkit for the Foundations of Computing Systems course based on a custom **32-bit RISC-V-inspired ISA** Assembler,Disassembler,Cpu-simulator,VM emulator.
- Using **Rocq** to explore formal verification and reasoning about the correctness of components of the system.
- Developing tools infrastructure including instruction representation, parsing, assembly, simulation, and verification components.
- *Tools:* OCaml, Rocq.
- Repository

Local LLM Debate System for Hallucination Reduction *Personal Project*

- Designed a local multi-agent LLM framework using **LangGraph**, where independent agents answer, critique, revise, and evaluate responses through multiple stages.
- Built a resource-routing pipeline that dynamically selects between **web search, RAG, local document retrieval, and Python tools** for evidence-grounded reasoning.
- Developed an event-driven architecture with a thread-safe event bus and real-time monitoring of routing decisions, retrieved evidence, debate rounds, and model outputs.
- *Tools:* Python, LangGraph, LangChain, Ollama, Qwen, ChromaDB, HuggingFace Embeddings, Rich, Pandas.
- GitHub Repository

Sheet2Slide - Agentic workflow *Personal Project*

- Built a multi-stage agentic AI pipeline using **LangGraph** that transforms structured spreadsheet data into presentation-ready reports.
- Engineered structured-output LLM agents for analytical reasoning while delegating statistical computation and visualization to deterministic Python tools.
- Implemented validation and fallback strategies to improve the reliability of generated analytical outputs.
- *Tools:* Python, LangGraph, Ollama, LangChain, Pandas, Matplotlib, Python-PPTX, Pydantic.
- GitHub Repository

Distributed Rate Limiting System with Redis and Lua *Personal Systems Project*

- Implemented Token Bucket, Leaky Bucket, Fixed Window, Sliding Window Log, and Sliding Window Counter algorithms.
- Built atomic **Redis + Lua** implementations to maintain correctness under concurrent and distributed workloads.
- Developed CI-based benchmarks to compare algorithmic performance under simulated traffic patterns.
- *Tools:* Python, Redis, Lua, GitHub Actions.
- GitHub Repository

TECHNICAL SKILLS

- **Programming Languages:** OCaml, Python, C, C++, JavaScript, SQL, Rocq
- **AI / ML:** PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, LangGraph, LangChain, Ollama, HuggingFace Embeddings, SHAP, LIME, CAM, Pandas, NumPy, Matplotlib
- **Software Engineering:** Git, GitHub Actions, Linux, REST APIs, Redis, Lua, Testing
- **Web Development:** FastAPI, Flask, React, JavaScript, HTML, CSS, PostgreSQL

ACHIEVEMENTS & LEADERSHIP

- **Inter-IIT Tech Meet 13.0:** Secured **8th position nationally** in the ISRO Problem Statement track.
- **Subsystem Lead - Avionics, Rocketry Team:** Lead the avionics subsystem for the INSPACE Model Rocketry competition, coordinating hardware/software integration and subsystem tasks. **Secured a top 15 position nationally**.
- **Core Team Member, NSS IIT Palakkad:** Coordinated and executed social service events on campus.